INTERNAL RELAY / STATE BIT ALLOCATION

M Address	Symbolic Name	State / Function	Set Condition	Reset Condition
M0001	SAFETY_READY	P00 ON and P07 ON	Safety Healthy	P00 OFF or P07 OFF
M0002	START_REQUEST	Start Command Latch	P01 Rising Edge	Cycle Start / P02
M0003	STOP_REQUEST	Normal Stop Latch	P02 Rising Edge	Return to S10
M0010	AUTO_MODE	Auto Mode Active	Start Request	Stop / Fault / Safety Loss
M0011	MANUAL_MODE	Manual / Setup Mode Active	D0053 = 1 or Manual Mode Select	D0053 = 0 / Auto Mode Select / Safety Loss
M0020	ROUTE_PASS	PASS Route Selected	M0109 ST_S81P Active	M0113 / M0003 / M0120
M0021	ROUTE_FAIL	FAIL Route Selected	M0110 ST_S81F Active	M0113 / M0003 / M0120
M0022	ROUTE_RETEST	RETEST Route Selected	M0111 ST_S81R Active	M0113 / M0003 / M0120
M0100	ST_S00	Safety Not Ready	Safety Loss	Safety Ready
M0101	ST_S10	Ready / Idle	Initialize Complete	Start Request
M0102	ST_S20	Wait Infeed	P01 Start	P04 ON
M0103	ST_S30	Transfer to Test	P04 ON	P05 ON / T0001
M0104	ST_S40	Clamp Close Command	P05 ON	Next Scan
M0105	ST_S50	Wait Clamp Closed	Clamp Command Issued	P08 ON / T0002
M0106	ST_S60	Test Execution	P08 ON	D0021 ON / T0003
M0107	ST_S70	Clamp Release	Test Complete	P08 OFF / T0004
M0108	ST_S80	Result Decision	Clamp Released	Result Routed
M0109	ST_S81P	PASS Gate Command	D0020 = 1	P09 ON / T0005
M0110	ST_S81F	FAIL Gate Command	D0020 = 2	P0A ON / T0005
M0111	ST_S81R	RETEST Gate Command	D0020 = 3	P0B ON / T0005
M0112	ST_S90	Transfer Outfeed	Gate Position Confirmed	P06 ON / T0006
M0113	ST_S100	Gate Return Home	P06 ON	Gate Feedback OFF
M0114	ST_S110	Cycle Complete	All Gates Home	Next Scan
M0120	ST_SF0	Fault / Alarm Hold	Any Process Fault	P03 Reset + Safety Ready

TIMER / ALARM FUNCTION REFERENCE

Tag	Function	Trigger State	Fault Action	Reference
T0001	Transfer to Test Timeout	S30	ALM-101 / SF0	DWG 012
T0002	Clamp Close Timeout	S50	ALM-102 / SF0	DWG 012
T0003	Test Complete Timeout	S60	ALM-103 / SF0	DWG 012
T0004	Clamp Release Timeout	S70	ALM-105 / SF0	DWG 012
T0005	Gate Position Timeout	S81-P/F/R	ALM-106 / SF0	DWG 012
T0006	Outfeed Timeout	S90	ALM-107 / SF0	DWG 012
T0007	MCU Communication Watchdog	S30~S90	ALM-108 / SF0	DWG 012
ALM-001	Safety Not Ready	Any State	S00 / P28 ON	DWG 012
ALM-101~108	Process Fault	Defined States	SF0 / P28 / P29	DWG 012

[GENERAL NOTES]

[1] PLC LADDER SHALL EXECUTE SAFETY STATUS MONITOR, STOP REQUEST, AND FAULT INTERLOCK NETWORKS BEFORE AUTO SEQUENCE OUTPUT NETWORKS.

[2] ONLY ONE AUTOMATIC SEQUENCE STATE BIT M0100 THROUGH M0114 SHALL BE ACTIVE AT ANY TIME.

[3] M0120 ST_SF0 FAULT HOLD SHALL FORCE PROCESS OUTPUTS P20 THROUGH P25 OFF AND COMMAND P28 RED LAMP / P29 BUZZER AS DEFINED ON DWG 012.

[4] PLC SHALL EVALUATE D0020 TEST RESULT CODE ONLY WHEN D0021 TEST COMPLETE FLAG = 1.

[5] PLC SHALL RESET D0022 TEST REQUEST FLAG ON EXIT FROM S60 OR WHEN FAULT STATE M0120 IS ACTIVE.

[6] P23, P24, AND P25 SHALL BE MUTUALLY EXCLUSIVE.

[7] TIMER AND ALARM SETTING DETAILS: SEE DWG 012.

[8] HARDWIRED SAFETY CIRCUIT DETAILS: SEE DWG 006.

PLC-01 / MCU-01 REGISTER INTERFACE MAP

Register	Signal	Direction	PLC Use	MCU Use	Reference
D0020	TEST_RESULT_CODE	MCU → PLC	Result Routing	Write PASS/FAIL/RETEST	DWG 011
D0021	TEST_COMPLETE_FLAG	MCU → PLC	Result Valid Check	Set After Result Valid	DWG 011
D0022	TEST_REQUEST_FLAG	PLC → MCU	Request Test	Start Test Evaluation	DWG 011
D0023	MCU_HEARTBEAT_COUNTER	MCU → PLC	Communication Watchdog	Increment Every 500 ms	DWG 012

NO.	DATE	REVISION
A	2026-09-10	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

PLC LADDER PROGRAM STRUCTURE

DRAWN BY: J. H. KO
DATE: 2026-09-10
DWG NO: 013
SHEET NO: 2 OF 3